

# EE3621: Digital Signal Processing

## Lecture 6: Inverse Z-Transform, System Response & Difference Equations (III B.Tech EEE)

### Lecture Notes & Comprehensive Worked Numericals

## 1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Motivation: Why Inverse Z-Transform? Uniqueness of ROC.
  - **05:00 – 12:00 (7 mins):** Three Core Inversion Approaches (PFE, Power Series, Contour Integration).
  - **12:00 – 20:00 (8 mins):** Worked Numericals: Real Exponentials ( $a^n$ ), Repeated Poles ( $na^n$ ), and Complex Conjugate Poles (Sinusoids/Cosines).
  - **20:00 – 27:00 (7 mins):** Convolution Theorem Equivalence (Time Domain Sum  $\leftrightarrow$  Z-Domain Product).
  - **27:00 – 34:00 (7 mins):** Solving Linear Constant-Coefficient Difference Equations (LC-CDE) with Initial Conditions.
  - **34:00 – 38:00 (4 mins):** Stability & Causality Analysis from Z-Plane Poles.
  - **38:00 – 40:00 (2 mins):** Summary & Review.
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## 2 Overview of the Inverse Z-Transform

The Inverse Z-transform maps  $X(z)$  and its ROC back to a discrete sequence  $x[n]$ :

$$x[n] = \mathcal{Z}^{-1}\{X(z)\} \quad (1)$$

Specification of the ROC is critical to ensure uniqueness.

## 3 The Three Inversion Methods

1. **Partial Fraction Expansion (PFE):** Decomposes  $\frac{X(z)}{z}$  into first/second-order pole components:

$$\frac{X(z)}{z} = \sum_{k=1}^N \frac{A_k}{z - p_k}, \quad A_k = \left[ (z - p_k) \frac{X(z)}{z} \right]_{z=p_k} \quad (2)$$

2. **Power Series Expansion (Long Division):** Direct polynomial long division:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \dots + x[-1]z^1 + x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots \quad (3)$$

3. **Contour Integration (Cauchy's Residue Theorem):**

$$x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1}dz = \sum \text{Residues of } [X(z)z^{n-1}] \text{ inside } C \quad (4)$$

## 4 Worked Numerical 1: Comparing All 3 Inversion Approaches

Given  $X(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}} = \frac{z^2}{(z-1)(z-0.5)}$ :

- **PFE Decomposition:**  $\frac{X(z)}{z} = \frac{2}{z-1} - \frac{1}{z-0.5} \implies X(z) = \frac{2}{1-z^{-1}} - \frac{1}{1-0.5z^{-1}}$ .
- **Causal Sequence** ( $|z| > 1$ ):  $x[n] = [2(1)^n - (0.5)^n]u[n] \implies \{1.0, 1.5, 1.75, 1.875, \dots\}$ .
- **Anti-Causal Sequence** ( $|z| < 0.5$ ):  $x[n] = [(0.5)^n - 2]u[-n-1] \implies \{\dots, 14, 6, 2, 0\}$ .
- **Two-Sided Sequence** ( $0.5 < |z| < 1$ ):  $x[n] = -(0.5)^n u[n] - 2u[-n-1]$ .

All three methods (PFE, polynomial long division, and residue integration) yield identical analytical formulas and sample values.

## 5 Worked Numerical 2: Complex Conjugate Poles (Sinusoids & Damped Cosines)

Given  $X(z) = \frac{z^2 - \frac{\sqrt{2}}{2}z}{z^2 - \sqrt{2}z + 1}$  with causal ROC  $|z| > 1$ : The poles lie on the unit circle at  $p_{1,2} = e^{\pm j\pi/4}$  ( $r = 1, \omega_0 = \pi/4$ ). Using the standard cosine pair  $\mathcal{Z}\{r^n \cos(\omega_0 n)u[n]\} = \frac{z(z-r \cos \omega_0)}{z^2 - 2r \cos \omega_0 z + r^2}$ :

$$x[n] = \cos\left(\frac{\pi}{4}n\right)u[n] \implies \{1.0, 0.707, 0.0, -0.707, -1.0, \dots\} \quad (5)$$

## 6 Worked Numerical 3: Repeated Poles ( $a^n$ and $na^n$ Ramp Modes)

Given  $X(z) = \frac{z^2}{(z-0.5)^2}$  with causal ROC  $|z| > 0.5$ :

$$\frac{X(z)}{z} = \frac{z}{(z-0.5)^2} = \frac{1.0}{z-0.5} + \frac{0.5}{(z-0.5)^2} \quad (6)$$

Multiplying by  $z$  and applying inverse pairs  $\frac{z}{z-a} \leftrightarrow a^n u[n]$  and  $\frac{az}{(z-a)^2} \leftrightarrow na^n u[n]$ :

$$x[n] = (0.5)^n u[n] + n(0.5)^n u[n] = (n+1)(0.5)^n u[n] \quad (7)$$

## 7 Convolution Theorem Equivalence: Time-Domain Sum $\leftrightarrow$ Z-Domain Product

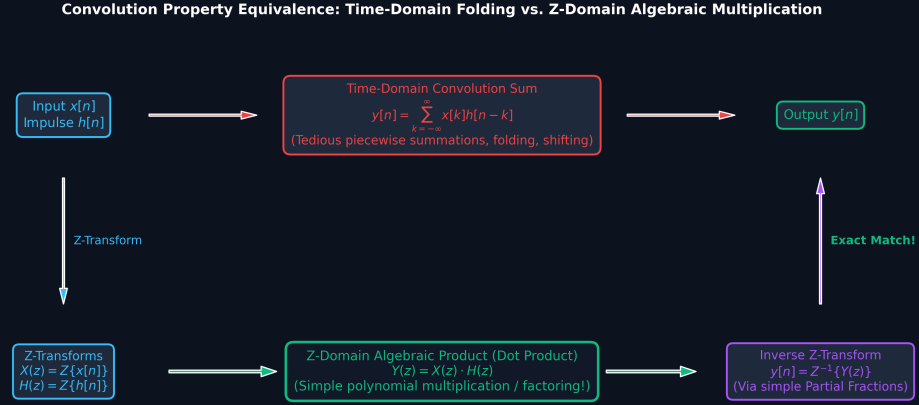


Figure 1: Convolution Property Equivalence: Time-Domain Folding vs. Z-Domain Algebraic Multiplication.

Given input  $x[n] = (0.5)^n u[n]$  and impulse response  $h[n] = (0.8)^n u[n]$ :

- **Time-Domain Sum:**  $y[n] = \sum_{k=0}^n (0.5)^k (0.8)^{n-k} = \frac{8}{3} (0.8)^n - \frac{5}{3} (0.5)^n, n \geq 0.$
- **Z-Domain Product:**  $Y(z) = X(z)H(z) = \frac{z^2}{(z-0.5)(z-0.8)} = \frac{8/3z}{z-0.8} - \frac{5/3z}{z-0.5}.$
- **Inverse Z Output:**  $y[n] = \left[ \frac{8}{3} (0.8)^n - \frac{5}{3} (0.5)^n \right] u[n].$

The Z-domain method replaces complex piecewise sliding summations with simple polynomial multiplication.

## 8 Solving Linear Difference Equations (LCCDE) via Z-Transform

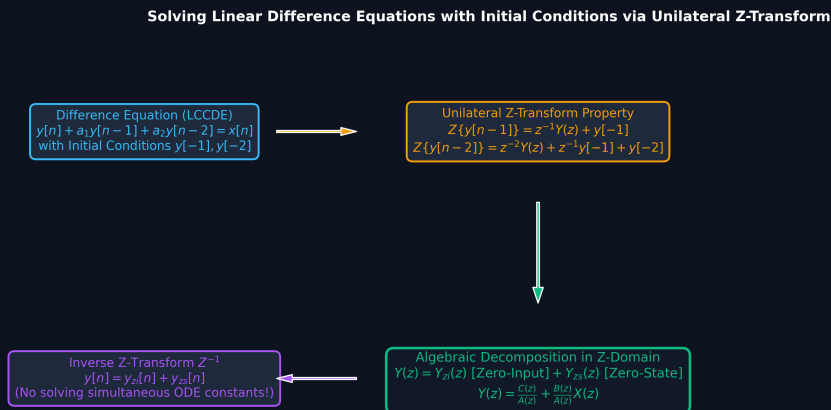


Figure 2: Solving Linear Difference Equations with Initial Conditions via Unilateral Z-Transform.

### 8.1 Worked Problem with Initial Conditions

Solve  $y[n] - 0.7y[n-1] + 0.1y[n-2] = x[n]$  for  $n \geq 0$ , with input  $x[n] = u[n]$  and initial conditions  $y[-1] = 2, y[-2] = 1$ .

**Step 1: Unilateral Z-Transform Property**

$$Y(z)(1 - 0.7z^{-1} + 0.1z^{-2}) - 0.7(2) + 0.1(2z^{-1} + 1) = X(z) \quad (8)$$

$$Y(z) = \underbrace{\frac{1.3 - 0.2z^{-1}}{1 - 0.7z^{-1} + 0.1z^{-2}}}_{Y_{zi}(z) \text{ (Zero-Input Response)}} + \underbrace{\frac{1}{1 - 0.7z^{-1} + 0.1z^{-2}}X(z)}_{Y_{zs}(z) \text{ (Zero-State Response)}} \quad (9)$$

**Step 2: Zero-Input Response**

$$Y_{zi}(z) = \frac{1.5z}{z - 0.5} - \frac{0.2z}{z - 0.2} \implies y_{zi}[n] = [1.5(0.5)^n - 0.2(0.2)^n] u[n] \quad (10)$$

**Step 3: Zero-State Response with  $X(z) = \frac{z}{z-1}$**

$$Y_{zs}(z) = \frac{2.5z}{z-1} - \frac{5/3z}{z-0.5} + \frac{1/6z}{z-0.2} \implies y_{zs}[n] = \left[ 2.5 - \frac{5}{3}(0.5)^n + \frac{1}{6}(0.2)^n \right] u[n] \quad (11)$$

**Step 4: Total Response  $y[n] = y_{zi}[n] + y_{zs}[n]$**

$$\mathbf{y[n]} = \left[ \mathbf{2.5} - \frac{\mathbf{1}}{\mathbf{6}}(\mathbf{0.5})^{\mathbf{n}} - \frac{\mathbf{1}}{\mathbf{30}}(\mathbf{0.2})^{\mathbf{n}} \right] \mathbf{u[n]} \quad (12)$$

Verified:  $y[0] = 2.3$  and  $y[1] = 2.41$ , matching difference recursion exactly.

## 9 Z-Plane Pole Locations & Stability Analysis

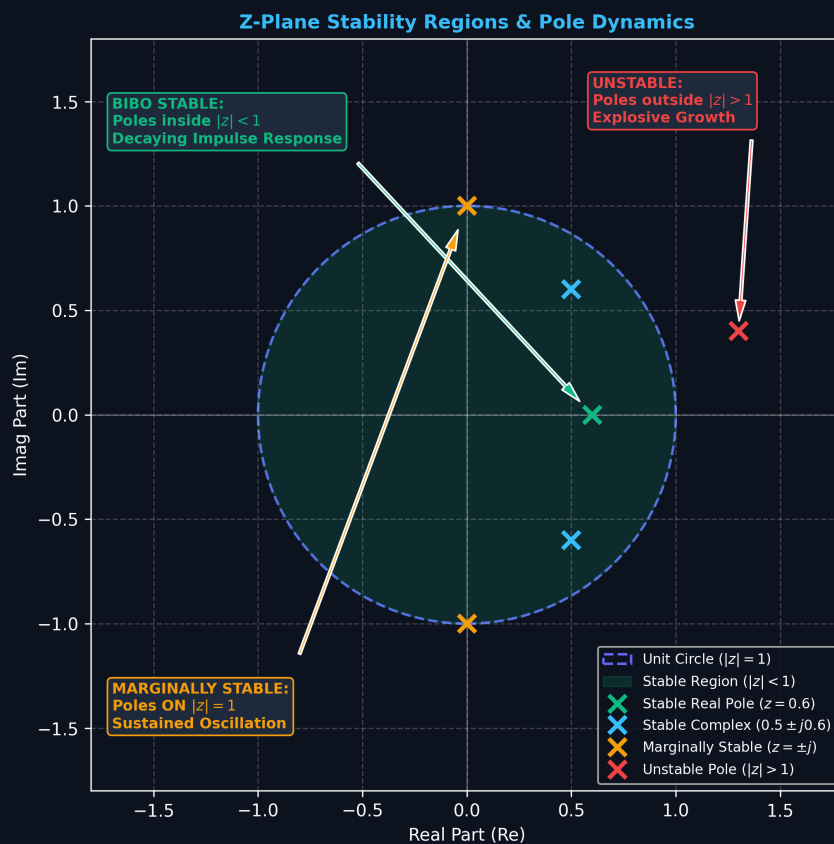


Figure 3: Z-Plane Stability Regions & Pole Dynamics: Stable ( $|z| < 1$ ), Marginally Stable ( $|z| = 1$ ), Unstable ( $|z| > 1$ ).

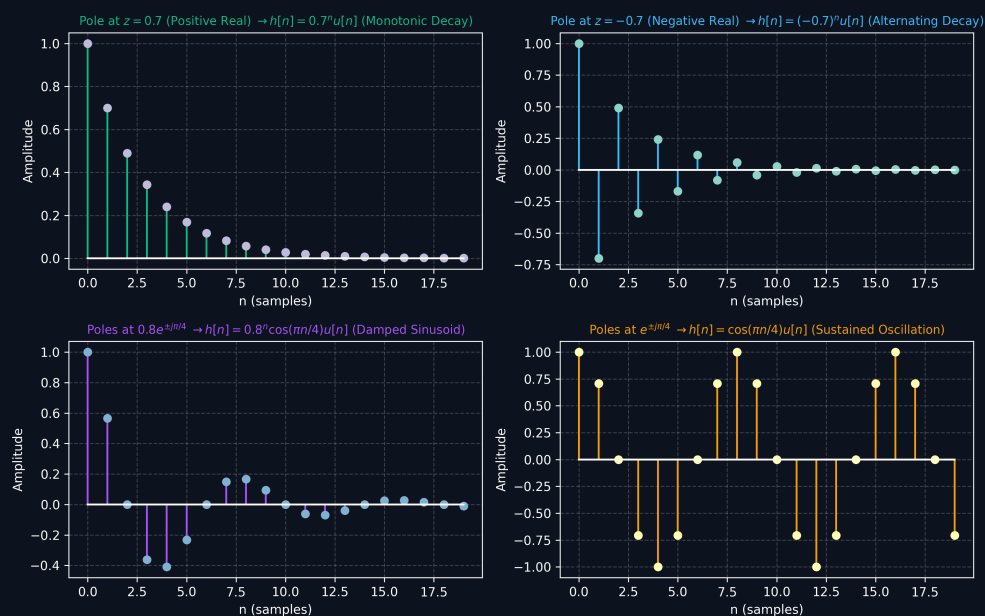


Figure 4: Time-Domain Impulse Response Modes corresponding to Pole Locations in the Complex Z-Plane.



Figure 5: Partial Fraction Expansion: Synthesizing Complete Impulse Response as Superposition of Modes.

- **Causality:** ROC of  $H(z)$  must be the exterior of the outermost pole:  $|z| > p_{max}$ .
- **BIBO Stability:** ROC of  $H(z)$  must include the unit circle ( $|z| = 1$ ).
- **Causal and Stable:** All poles must lie strictly inside the unit circle ( $|p_k| < 1$ ).